Learning Healthy Anatomy to Find Tumors: Comparing VAEs and GANs for Unsupervised Anomaly Detection in Brain MRI

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Problem statement and motivation

We ask whether a generative model that has only ever seen healthy brains can reliably detect and localize brain tumors, and whether a variational autoencoder (VAE) or a generative adversarial network (GAN) does this better.

The problem with the usual approach is that most deep learning tools for medical imaging are supervised, so they learn from thousands of scans where a radiologist has hand-drawn the outline of every tumor. That labeling is slow, expensive and requires experts. Supervised models also only recognize the specific diseases they were trained on, so a rare or unusual abnormality can slip past them.

Our approach is to learn "normal," then flag what is different. Healthy brain scans are much easier to collect than labeled tumor scans. If a model learns what healthy anatomy looks like well enough to recreate it, then when it is shown a scan with a tumor, it will try to "repair" the image back to a healthy-looking version. The places where its reconstruction differs most from the real scan should be where the abnormality is. This is called unsupervised anomaly detection, because no tumor labels are needed for training.

We will compare VAEs and GANs because both are generative models, but they learn in different ways and have known tradeoffs. VAEs are stable to train but tend to produce blurry images, which can create false alarms at fine edges. GANs produce sharper images but are harder to train and can miss parts of the data. Prior studies report mixed results depending on dataset and preprocessing, so a controlled side-by-side comparison on the same public data is useful.

Our resulting model could act as a "second reader" that highlights suspicious regions for a radiologist, including abnormalities no one trained it to look for. Our results will show how close these methods come to that goal, and where they still fail.

Literature review: Prior work establishes both model families as anomaly detectors for brain MRI, but finds no clear winner, which motivates our controlled comparison.

VAEs and GANs

Kingma and Welling [1] introduced the VAE, which compresses an image into a small set of numbers (a latent code) and decodes it back, trained so that the latent space is smooth and well-organized. Goodfellow et al. [2] introduced the GAN, in which a generator creates images and a discriminator tries to tell them apart from real ones, pushing the generator toward realistic output. Radford et al. [3] showed that a convolutional design (DCGAN) makes GANs much more stable on images; we will use DCGAN-style layers in both our models.

GANs for anomaly detection:

Schlegl et al. [4] proposed AnoGAN, the first GAN used to find anomalies in medical images (retinal scans). A GAN trained on healthy images is searched for the closest healthy-looking version of a test image, and the difference is the anomaly map. Its weakness is speed, since that search runs separately for every image. The follow-up f-AnoGAN [5] adds an encoder that maps an image straight

to its latent code, making detection fast enough for practical use. We will use f-AnoGAN as our GAN method.

Autoencoders on brain MRI:

Baur et al. [6] combined a VAE with an adversarial loss to segment multiple sclerosis lesions in brain MRI without labels, showing that sharper reconstructions reduce false alarms. Their later comparative study [7] benchmarked many autoencoder and GAN variants on the same brain MRI data. A key finding was that results depend heavily on preprocessing and on how the anomaly map is post-processed, and simple methods were often competitive with complex ones. This shapes our plan: we will keep preprocessing identical across models and report post-processing choices explicitly.

Most comparisons use private hospital data, which makes them hard to reproduce. We will run a VAE versus f-AnoGAN comparison entirely on public, freely downloadable data (IXI and the Medical Segmentation Decathlon [8]), with shared code and preprocessing, so the result can be checked by anyone.

Datasets:

We will use two public, de-identified datasets that download directly: IXI for training on healthy brains and the Medical Segmentation Decathlon brain tumor task for testing. We will follow both CC BY-SA licenses by citing the sources.

IXI (training): MRI scans of about 600 healthy adult volunteers, all scans are healthy, available sequences are T1, T2, PD, and others, we will use T2, License: CC BY-SA 3.0. This will teach models what normal anatomy looks like.

MSD Task 01 BrainTumour (testing): 484 MRI scans of patients with glioma (a common brain tumor), derived from the BraTS challenge, labeled as expert tumor outlines for every scan, available sequences are FLAIR, T1, contrast-enhanced T1, T2, we will use T2, license is CC BY-SA 4.0. This will measure how well models find and outline tumors.

T2-weighted MRI

T2 is the only sequence both datasets share, and tumors and the swelling around them show up bright on T2, which gives the models a clear signal to detect.

Preprocessing

IXI and MSD were collected on different scanners, so without careful matching a model might flag "different hospital" instead of "tumor." We will apply the same pipeline to both:

1. Skull-strip each scan so only brain tissue remains.
2. Register (align) every scan to a common brain template so anatomy lines up.
3. Normalize intensities so brightness values are on the same scale.
4. Extract 2D axial slices from the middle of the brain, skipping near-empty top and bottom slices.
5. Resize slices to 128×128 pixels (64×64 for early fast experiments).

IXI is split 80/10/10 into training, validation and held-out healthy test sets by subject, so no person appears in more than one split. MSD scans are used only for validation of the detection threshold (a small subset) and final testing.

We will train three models on healthy IXI slices, all with DCGAN-style convolutional layers:

- VAE with a 128–256 dimensional latent code.
- f-AnoGAN [5]: a GAN plus an encoder into its latent space.
- Plain autoencoder baseline, to test whether generative modeling adds value.

How detection works: For each test slice, the model produces a healthy-looking reconstruction. We subtract it from the original, pixel by pixel, to get an anomaly map: bright where the model failed to reproduce the image. We lightly smooth the map, keep only pixels inside the brain, and apply a threshold chosen on a small validation subset to produce a predicted tumor outline.

We report Dice overlap with expert outlines, pixel-level AUROC and AUPRC, slice-level AUROC (does a slice contain a tumor?), false-positive rate on healthy IXI test slices, and side-by-side visual examples.

References

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